

3.6 The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

3.7 The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

- (a) less than 120 hours;
- (b) between 50 and 100 hours.

Exercise Question 3.6:

Given

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3} & , x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

a - at least 200 days.

b - anywhere from 80 to 120 days.

Sol:

$$\begin{aligned} \text{a- } P(X \geq 200) &= \int_{-\infty}^{+\infty} f(x) dx = \int_{200}^{\infty} \frac{20,000}{(x+100)^3} dx \quad \xrightarrow{\text{by power rule}} \\ &= 20,000 \int_{200}^{\infty} (x+100)^{-3} dx \\ &= 20,000 \left[\frac{(x+100)^{-3+1}}{-3+1} \right]_{200}^{\infty} \\ &= 20,000 \left[\frac{(x+100)^{-2}}{-2} \right]_{200}^{\infty} = -10,000 \left[\frac{1}{(x+100)^2} \right]_{200}^{\infty} \\ &= -10,000 \left[\frac{1}{(\infty+100)^2} - \frac{1}{(200+100)^2} \right] = -10,000 \left[0 - \frac{1}{(300)^2} \right] \\ &= \frac{10,000}{90,000} = 0.1111 \end{aligned}$$

$$\begin{aligned} \text{b- } P(80 \leq X \leq 200) &= \int_{-\infty}^{+\infty} f(x) dx = \int_{80}^{200} \frac{20,000}{(x+100)^2} dx \\ &= -10,000 \left[\frac{1}{(x+100)^2} \right]_{80}^{200} \\ &= -10,000 \left[\frac{1}{(200+100)^2} - \frac{1}{(80+100)^2} \right] \\ &= -10,000 \left[\frac{1}{(300)^2} - \frac{1}{(180)^2} \right] = -10,000 \left[0.001111 - 0.003086 \right] \\ &= 0.1975 \end{aligned}$$

$$\text{Also, } P(80 \leq X \leq 120) = 0.1020$$

Exercise question 3.7:

Given

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

a- Less than 120 hours

b- between 50 and 100 hours

Given

100 hours = 1 unit

Sol: $P(X < 120)$

$$P(X < 1.2) = \int_0^1 f(x) dx + \int_1^{1.2} f(x) dx$$

$$= \int_0^1 x dx + \int_1^{1.2} (2-x) dx$$

$$= \left. \frac{x^2}{2} \right|_0^1 + \int_1^{1.2} 2 dx - \int_1^{1.2} x dx$$

$$= \left. \frac{x^2}{2} \right|_0^1 + 2x \Big|_1^{1.2} - \left. \frac{x^2}{2} \right|_1^{1.2}$$

$$= \left[\frac{1}{2} - 0 \right] + 2[1.2 - 1] - \frac{1}{2} [(1.2)^2 - (1)^2] = \boxed{0.68}$$

$$= \frac{1}{2} + 0.4 - \frac{1}{2}(0.44) = \frac{1}{2} + 0.4 - 0.22$$

b- $P(50 < X < 100) = P(0.5 < X < 1) = \int_{-\infty}^{+\infty} f(x) dx$

$$= \int_{0.5}^1 f(x) dx = \left. \frac{x^2}{2} \right|_{0.5}^1$$

$$= \frac{1}{2} - \frac{(0.5)^2}{2} = \boxed{0.375}$$

1-

⇒ Mean and Variance of Continuous Random variable

$$\mu = E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{where } E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx$$

2-

⇒ Mean and Variance of function of Continuous Random variable.

Let $g(x)$ be a function of Random variable 'x'

then

$$\mu = E[g(x)] = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

$$\text{Var}(g(x)) = E[g(x)]^2 - [E(g(x))]^2$$

where

$$E[g(x)]^2 = \int_{-\infty}^{+\infty} [g(x)]^2 f(x) dx$$

See example 4.10 for 1

See example 4.5 & 4.12 for 2

4.10

$$f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find mean and variance of x .

Sol:

$$\bullet \mu = E(x) = \int_{-\infty}^{+\infty} x f(x) dx = \int_1^2 x \cdot 2(x-1) dx$$

$$= 2 \int_1^2 (x^2 - x) dx = 2 \left[\frac{x^3}{3} \right]_1^2 - 2 \left[\frac{x^2}{2} \right]_1^2$$

$$= \frac{2}{3} [2^3 - 1^3] - [2^2 - 1^2] = \frac{14}{3} - 3 = \boxed{\frac{5}{3}}$$

$$\bullet \text{var}(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_1^2 x^2 \cdot 2(x-1) dx$$

$$= 2 \int_1^2 (x^3 - x^2) dx = 2 \left[\frac{x^4}{4} \right]_1^2 - \frac{2x^3}{3} \Big|_1^2$$

$$= \frac{1}{2} [2^4 - 1^4] - \frac{2}{3} [2^3 - 1^3] = \frac{15}{2} - \frac{14}{3} = 2.8333 = \boxed{\frac{17}{6}}$$

4.5 & 4.12

Given $f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$

1- Find expected value of $g(x) = 4x + 3$

2- Find Variance of $g(x) = 4x + 3$

$$\therefore E(g(x)) = \int_{-\infty}^{+\infty} g(x)f(x) dx$$

Sol

1- $E(g(x)) = E(4x + 3) = \int_{-\infty}^{+\infty} (4x + 3) \frac{x^2}{3} dx$

$$= \int_{-1}^2 (4x + 3) \frac{x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx$$

$$= \frac{1}{3} \left[\frac{4x^4}{4} \Big|_{-1}^2 + \frac{3x^3}{3} \Big|_{-1}^2 \right] = \frac{1}{3} \left[x^4 \Big|_{-1}^2 + x^3 \Big|_{-1}^2 \right]$$

$$= \frac{1}{3} \left[\{2^4 - (-1)^4\} + \{2^3 - (-1)^3\} \right]$$

$$= \frac{1}{3} [(16 - 1) + (8 + 1)] = \frac{1}{3} [15 + 9] = \frac{24}{3} = 8$$

2- $\text{Var}(g(x)) = E(g(x)^2) - [E(g(x))]^2$

$$\rightarrow E(g(x)^2) = \int_{-\infty}^{+\infty} g(x)^2 f(x) dx$$

$$= \int_{-1}^2 (4x + 3)^2 \frac{x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (16x^2 + 9 + 24x) x^2 dx$$

$$= \frac{1}{3} \int_{-1}^2 (16x^4 + 9x^2 + 24x^3) dx = \frac{1}{3} \left[\frac{16x^5}{5} \Big|_{-1}^2 + \frac{9x^3}{3} \Big|_{-1}^2 + \frac{24x^4}{4} \Big|_{-1}^2 \right]$$

$$= \frac{1}{3} \left[\left[\frac{16(2^5)}{5} - \frac{16(-1)^5}{5} \right] + (3(2)^3 - 3(-1)^3) + [6(2)^4 - 6(-1)^4] \right]$$

$$= \frac{1}{3} [(102.4 + 3.2) + (24 + 3) + (96 - 6)] = \frac{1}{3} (105.6 + 27 + 90) = \frac{222.6}{3} = 74.2$$

$$= \frac{1}{3} [105.6 + 27 + 90] = 74.2$$

so)

$$\text{var}(g(x)) = 74.2 - 8^2 = \boxed{10.2}$$